



GIRRAWEEEN HIGH SCHOOL

2004

YEAR 11 HALF YEARLY EXAMINATIONS

TASK 1

Mathematics Extension 1

General Instructions

- Working time – 90 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work

Total marks – 108

- Attempt Questions 1 – 5
- Questions are NOT of equal value

Total marks – 113

Attempt Questions 1 – 5

Questions are NOT of equal value

Answer each question on a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

Marks

Question 1 (20 marks) Use a separate piece of paper

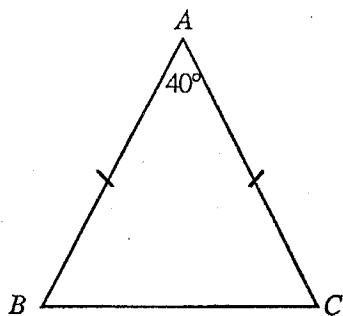
- a) If $f(x) = x^2 + x$ and $g(x) = x^2 + 3x + 1$, find;
- (i) $f(1)$ 1
 - (ii) $g(-2)$ 1
 - (iii) $f(4) - g(3)$ 2
 - (iv) $f(x+1)$ 2
- b) Given that $f(x) = \begin{cases} x^2 & , x > 2 \\ \frac{1}{x+1} & , -1 < x \leq 2 \\ 5 & , x \leq -1 \end{cases}$, find;
- (i) $f(3) + f(-1)$ 2
 - (ii) the domain of $f(x)$ 1
 - (iii) the range of $f(x)$ 1
- c) In your own words describe what is meant by “function” when referring to number plane graphs. 2
- d) Write down the domain and range of;
- (i) $y = 2x^2 - 5$ 2
 - (ii) $y = \frac{4}{x-2}$ 2
 - (iii) $y = \sqrt{16 - x^2}$ 2
 - (iv) $y = \frac{5}{x^2 + 9}$ 2

Question 2 (24 marks) Use a separate piece of paper

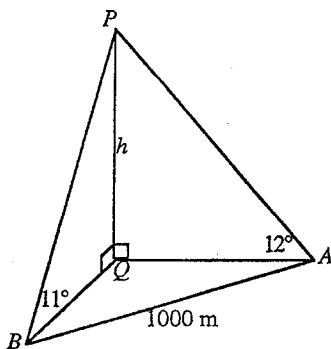
- a) Solve the following inequalities;
- (i) $\frac{2}{x-3} \geq 4$ 3
 - (ii) $\frac{1}{x} < \frac{1}{x+1}$ 3

Question 2...continued.**Marks**b) Solve for θ , correct to the nearest degree where necessary, where $0^\circ \leq \theta \leq 360^\circ$

- (i) $\cos^2 \theta = \cos \theta$ 3
- (ii) $\sqrt{3} \sin \theta - 3 \cos \theta = 0$ 3
- (iii) $2 \sin^2 \theta - 5 \sin \theta + 2 = 0$ 4
- (iv) $\sin 3\theta = \frac{1}{2}$ 4
- (v) $4 \tan \theta + 1 - \tan^2 \theta = \sec^2 \theta$ 4

Question 3 (13 marks) Use a *separate* piece of papera) The vertical angle of an isosceles triangle is 40° and its area is 40 cm^2 .

- (i) Show that $AB = \sqrt{80 \operatorname{cosec} 40^\circ}$ 2
- (ii) Hence, or otherwise, calculate the length of BC , correct to one decimal place 3
- b) The angle of elevation of a tower PQ of height h metres, at a point A due East of it, is 12° . From another point B , the bearing of the tower is 051°T and the angle of elevation is 11° . The points A and B are 1000 metres apart and on the same level as the base Q of the tower.



- (i) Show that $\angle AQB = 141^\circ$ 2
- (ii) Consider $\triangle APQ$, show that $AQ = h \tan 78^\circ$ 2
- (iii) Find a similar expression for BQ 1
- (iv) Hence calculate h , correct to the nearest metre. 3

Question 4 (36 marks) Use a *separate* piece of paper

- a) The game of Yahtzee involves rolling five regular six sided dice.
- (i) How many ways can the dice land? 2
 - (ii) A Yahtzee is when all five dice show the same number. 2
What is the probability of rolling a Yahtzee?
- b) An eight person committee is to be chosen from six boys and six girls. In how many ways can this be done if;
- (i) there are no restrictions? 2
 - (ii) there must be an equal number of boys and girls on the committee? 2
 - (iii) the two school captains (one boy and one girl) must be on the committee? 2
- c) How many ways can the letters of the word PARRAMATTA be arranged if;
- (i) there are no restrictions? 2
 - (ii) an R must be at either end of the word? 2
 - (iii) the T's cannot be next to each other? 2
- d) To win Lotto, you choose 6 numbers out of a total of 45.
- (i) In how many ways can this be done? 2
 - (ii) What is the probability of winning first prize, (that is picking all six numbers correctly), if you have four different entries? 2
- e) Numbers are to be made from the digits 1 to 9, with no digit allowed more than once in the same number.
- (i) How many four digit numbers can be made? 2
 - (ii) How many of these four digit numbers would be odd? 2
 - (iii) What is the probability that a four digit number is divisible by 5? 3
 - (iv) How many numbers, (using any number of digits), less than 3000 can be made? 3
- f) Four men and four women are seated around a table
- (i) How many ways can they be seated? 2
 - (ii) What is the probability that the men and women alternate around the table? 2
 - (iii) What is the probability that two particular men do not sit next to each other? 2

Question 5 (15 marks) Use a *separate* piece of paper

a) Factorise $(a^2 - b^2)^2 - (a - b)^4$ completely 3

b) If θ is acute and $\sin \theta = \frac{1}{\sqrt{3}}$

(i) Show that $\frac{\tan \theta}{1 - \sec \theta} = -\sqrt{2} - \sqrt{3}$ 3

(ii) Find the value of this fraction when θ is obtuse. 2

c) Prove that $\frac{1 + \cot \theta}{\operatorname{cosec} \theta} - \frac{1 + \tan \theta}{\sec \theta}$ is independent of θ 3

d) Prove that $\frac{1 + \cos \theta}{1 - \cos \theta} = (\operatorname{cosec} \theta + \cot \theta)^2$ 4

Question 1 (20)

a) $f(x) = x^2 + x$ $g(x) = x^2 + 3x + 1$

(i) $f(1) = 1^2 + 1 = 2$ ①

(ii) $g(-2) = (-2)^2 + 3(-2) + 1 = -1$ ①

(iii) $f(4) - g(3) = 4^2 + 4 - [3^2 + 3(3) + 1] = 20 - 19 = 1$ ②

(iv) $f(x+1) = (x+1)^2 + (x+1) = x^2 + 2x + 1 + x + 1 = x^2 + 3x + 2$ ②

b) (i) $f(3) + f(-1) = 3^2 + 5 = 14$ ②

(ii) domain: all real x ①

(iii) range: $y \geq 0$ ①

c) A function is a relation such that for every value in the domain there exists no more than one value in the range ②

d) (i) domain: all real x ①
range: $y \geq -5$ ①

(ii) domain: all real x except $x=2$ ①
range: all real y except $y=0$ ①

(iii) domain: $16 - x^2 \geq 0$
 $x^2 \leq 16$
 $-4 \leq x \leq 4$ ①

range: $x=4, y=\sqrt{16-4^2}=0$
 $x=0, y=\sqrt{16-0^2}=4$
 $0 \leq y \leq 4$ ①

(iv) domain: all real x ①
range: $x^2 \geq 0$
 $x^2 + 9 \geq 9$
 $0 < y \leq \frac{5}{9}$ ①

Question 2 (24)

a) $\frac{2}{x-3} > 4$
 $x \neq 3$

$\frac{2}{x-3} = 4$
 $2 = 4x - 12$
 $4x = 14$
 $x = \frac{7}{2}$

$3 < x \leq \frac{7}{2}$ ③

(ii) $\frac{1}{x} < \frac{1}{x+1}$
 $x \neq 0, x \neq -1$
 $\frac{1}{x} - \frac{1}{x+1} < 0$
 $\frac{x+1-x}{x(x+1)} < 0$
 $\frac{1}{x(x+1)} < 0$
 $0 = 1$
no solutions

$-1 < x < 0$ ③

b) (i) $\cos^2 \theta = \cos \theta$
 $\cos^2 \theta - \cos \theta = 0$
 $\cos \theta (\cos \theta - 1) = 0$
 $\cos \theta = 0$ or $\cos \theta = 1$
 $\theta = 90^\circ, 270^\circ$ or $\theta = 0^\circ, 360^\circ$
 $\theta = 0^\circ, 90^\circ, 270^\circ, 360^\circ$ ③

(ii) $\sqrt{3} \sin \theta - 3 \cos \theta = 0$
 $\sqrt{3} \sin \theta = 3 \cos \theta$
 $\tan \theta = \sqrt{3}$
 $Q1, Q3$
 $\tan \alpha = \sqrt{3}$
 $\alpha = 60^\circ$
 $\theta = 60^\circ, 240^\circ$ ③

(iii) $2 \sin^2 \theta - 5 \sin \theta + 2 = 0$
 $2 \sin^2 \theta - 4 \sin \theta - \sin \theta + 2 = 0$
 $2 \sin \theta (\sin \theta - 2) - 1(\sin \theta - 2) = 0$
 $(\sin \theta - 2)(2 \sin \theta - 1) = 0$
 $\sin \theta = 2$ or $\sin \theta = \frac{1}{2}$
no solutions or $Q1, Q2$
 $\sin \alpha = \frac{1}{2}$
 $\alpha = 30^\circ$
 $\theta = 30^\circ, 150^\circ$
 $\theta = 30^\circ, 150^\circ$ ④

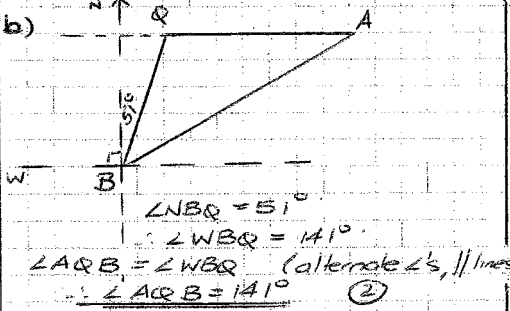
(iv) $\sin 3\theta = \frac{1}{2}$ $0 \leq 3\theta \leq 1080^\circ$
 $Q1, Q2$
 $\sin \alpha = \frac{1}{2}$
 $\alpha = 30^\circ$
 $3\theta = 30^\circ, 150^\circ, 390^\circ, 510^\circ, 750^\circ, 870^\circ$
 $\theta = 10^\circ, 50^\circ, 130^\circ, 170^\circ, 250^\circ, 290^\circ$ ④

(iv) $4 \tan \theta + 1 - \tan^2 \theta = \sec^2 \theta$
 $= 1 + \tan^2 \theta$
 $2 \tan^2 \theta - 4 \tan \theta = 0$
 $2 \tan \theta (\tan \theta - 2) = 0$
 $\tan \theta = 0$ or $\tan \theta = 2$
 $\theta = 0^\circ, 180^\circ, 360^\circ$ or $Q1, Q3$
 $\tan \alpha = 2$
 $\alpha = 63^\circ$
 $\theta = 63^\circ, 143^\circ$
 $\theta = 0^\circ, 63^\circ, 143^\circ, 180^\circ, 360^\circ$ ④

Question 3 (13)

(i) $\frac{1}{2} AB \cdot AC \sin 40^\circ = 40$
 $\frac{1}{2} AB^2 \sin 40^\circ = 40$
 $AB^2 = \frac{80}{\sin 40^\circ}$
 $AB = \sqrt{80 \csc 40^\circ}$ ②

(ii) $BC^2 = AB^2 + AC^2 - 2 \cdot AB \cdot AC \cos 40^\circ$
 $= 2AB^2 - 2AB^2 \cos 40^\circ$
 $= 2AB^2 (1 - \cos 40^\circ)$
 $= 160 \csc^2 40^\circ (1 - \cos 40^\circ)$
 $BC = 7.6 \text{ cm}$ ③



(iii) $\frac{AQ}{h} = \tan 78^\circ$
 $AQ = h \tan 78^\circ$ ②

(iv) $\frac{BQ}{h} = \tan 79^\circ$
 $BQ = h \tan 79^\circ$ ①

(v) $AB^2 = AQ^2 + BQ^2 - 2AQ \cdot BQ \cos Q$
 $1000^2 = h^2 \tan^2 78^\circ + h^2 \tan^2 79^\circ - 2h^2 \tan 78^\circ \tan 79^\circ \cos 141^\circ$
 $h^2 = \frac{1000^2}{\tan^2 78^\circ + \tan^2 79^\circ - 2 \tan 78^\circ \tan 79^\circ \cos 141^\circ}$
 $h = 108 \text{ m}$ ③

Question 4 (36)

a) (i) Ways = $\frac{6^5}{7776}$ ②
(ii) $P(\text{Yahtzee}) = \frac{6}{1296}$ ②

b) (i) Ways = $\frac{12C_8}{495}$ ②

(ii) Ways = $\frac{6C_4 \times 6C_4}{225}$ ②

(iii) Ways = $\frac{1 \times 10C_6}{210}$ ②

c) (i) Ways = $\frac{10!}{4!2!2!}$
 $= 37800$ ②

(ii) Ways = $1 \times \frac{8!}{4!2!}$
 $= 840$ ②

(iii) Ways T's together = $\frac{9!}{4!2!}$
 $= 7560$

$\therefore$ Ways T's not together = $37800 - 7560 = 30240$ ②

d) (i) Ways = $\frac{15C_6}{8145060}$ ②

(ii) $P(\text{win}) = \frac{4}{15C_6}$
 $= \frac{1}{2036265}$ ②

e) (i) 4 digit #'s = $\frac{9P_4}{3024}$ ②

(ii) odd 4 digit #'s = $\frac{5P_1 \times 8P_3}{1680}$ ②

(iii) $\div 5$ means # ends in 5
4 digit $\div 5 = \frac{1 \times 8P_3}{336}$
 $P(\div 5) = \frac{336}{3024}$
 $= \frac{1}{9}$ ③

$$\begin{aligned}
 \text{(iv) 1 digit} &= {}^9P_1 = 9 \\
 2 \text{ digit} &= {}^9P_2 = 72 \\
 3 \text{ digit} &= {}^9P_3 = 504 \\
 4 \text{ digit} &= 2 \times {}^8P_3 = 672
 \end{aligned}$$

$$\begin{aligned}
 \#5 &< 3000 = 9 + 72 + 504 + 672 \\
 &= \underline{1257} \quad (3)
 \end{aligned}$$

$$\text{(i) Ways} = 7! \quad (2)$$

$$\begin{aligned}
 \text{(ii) } P(\text{alternate}) &= \frac{3! \cdot 4!}{7!} \\
 &= \frac{1}{35} \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) } P(\text{not together}) &= 1 - P(\text{together}) \\
 &= 1 - \frac{2! \cdot 6!}{7!} \\
 &= 1 - \frac{2}{7} \\
 &= \frac{5}{7} \quad (2)
 \end{aligned}$$

Question 5 (15)

$$\begin{aligned}
 \text{i) } (a^2 - b^2)^2 - (a - b)^4 \\
 &= (a + b)^2 (a - b)^2 - (a - b)^4 \\
 &= (a - b)^2 [(a + b)^2 - (a - b)^2] \\
 &= (a - b)^2 (4ab) \\
 &= \underline{4ab(a - b)^2} \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } \frac{\tan \theta}{1 - \sec \theta} &= \frac{\frac{1}{\sqrt{2}}}{1 - \frac{\sqrt{2}}{2}} \\
 &= \frac{1}{\sqrt{2} - \sqrt{3}} \times \frac{\sqrt{2} + \sqrt{3}}{\sqrt{2} + \sqrt{3}} \\
 &= \frac{\sqrt{2} + \sqrt{3}}{\sqrt{2} - \sqrt{3}} \\
 &= \underline{-\sqrt{2} - \sqrt{3}} \quad (3)
 \end{aligned}$$

iii) if θ obtuse;

$$\begin{aligned}
 \frac{\tan \theta}{1 - \sec \theta} &= \frac{\frac{1}{\sqrt{2}}}{1 + \frac{\sqrt{2}}{2}} \\
 &= \frac{1}{\sqrt{2} + \sqrt{3}} \times \frac{\sqrt{2} - \sqrt{3}}{\sqrt{2} - \sqrt{3}} \\
 &= \frac{\sqrt{2} - \sqrt{3}}{\sqrt{2} + \sqrt{3}} \\
 &= \underline{\sqrt{2} - \sqrt{3}} \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } \frac{1 + \cot \theta}{\operatorname{cosec} \theta} &= \frac{1}{\sec \theta} \left(\frac{\sin \theta}{\cos \theta} \right) \\
 &= \frac{1 + \frac{\cos \theta}{\sin \theta}}{\frac{1}{\sin \theta}} = \frac{1 + \frac{\sin \theta}{\cos \theta}}{\frac{1}{\cos \theta}}
 \end{aligned}$$

$$\begin{aligned}
 &= \sin \theta + \cos \theta - (\cos \theta + \sin \theta) \\
 &= \underline{0} \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } \frac{1 + \cos \theta}{1 - \cos \theta} \times \frac{1 + \cos \theta}{1 + \cos \theta} \\
 &= \frac{1 + 2\cos \theta + \cos^2 \theta}{1 - \cos^2 \theta} \\
 &= \frac{1 + 2\cos \theta + \cos^2 \theta}{\sin^2 \theta} \\
 &= \operatorname{cosec}^2 \theta + 2 \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta} + \cot^2 \theta \\
 &= \operatorname{cosec}^2 \theta + 2 \cot \theta \operatorname{cosec} \theta + \cot^2 \theta \\
 &= \underline{(\operatorname{cosec} \theta + \cot \theta)^2} \quad (4)
 \end{aligned}$$